LAPUTA

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Individual Project；Full-Pipeline Stylized Scene Project: PCG + SD with Strong Technical Art Focus

This project draws inspiration from the "Laputa's cabin" in Hayao Miyazaki's Castle in the Sky. I translated 2D hand-painted aesthetics into a real-time 3D rendered environment while exploring the full-cycle technical implementation of Procedural Content Generation (PCG).

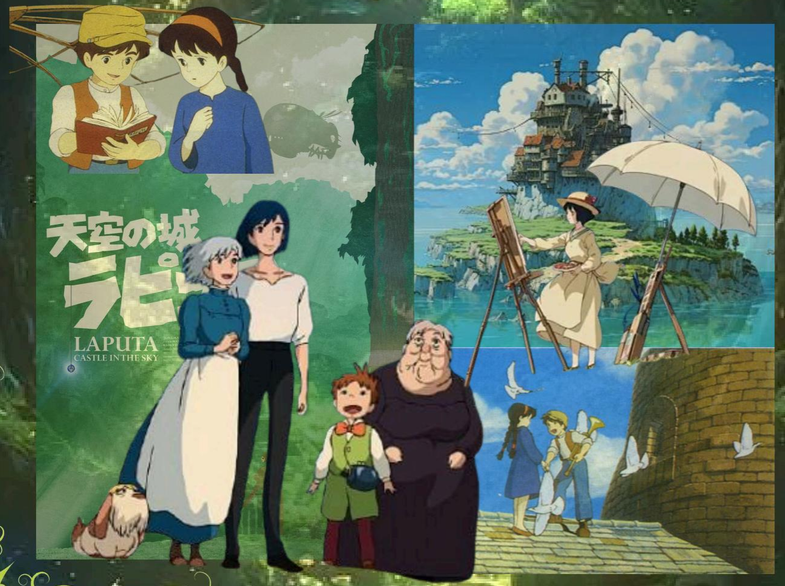
Using Substance Designer, I developed 3 material maps based on the original art. Additionally, with Houdini, I created parametric tools that enable one-click generation of stylized brick walls, spline-based stone wall creation, and layered mountain terrain via HeightField systems—greatly accelerating the efficiency of scene iteration.

Within Unreal Engine 5, I applied custom Toon Shaders, vertex-color materials, and Runtime Virtual Texturing (RVT) techniques to enhance the simplicity of colors and the abstraction of lighting, thus recreating Studio Ghibli's distinctive soft visual style.

Inspiration

WHY ?

This project originated from my long-time admiration for Hayao Miyazaki's works. When I saw this image of Laputa's home in Castle in the Sky, its palette reminded me of the tones I love in ICO, yet it was rendered in a stylized manner—which inspired me to give it a try. At the same time, I noticed that many of its assets involved foundational workflows created with Houdini HDAs, so I also wanted to use this as an oppottunity to practice hands-on Houdini development.



Stylized Material

Brick Wall Material

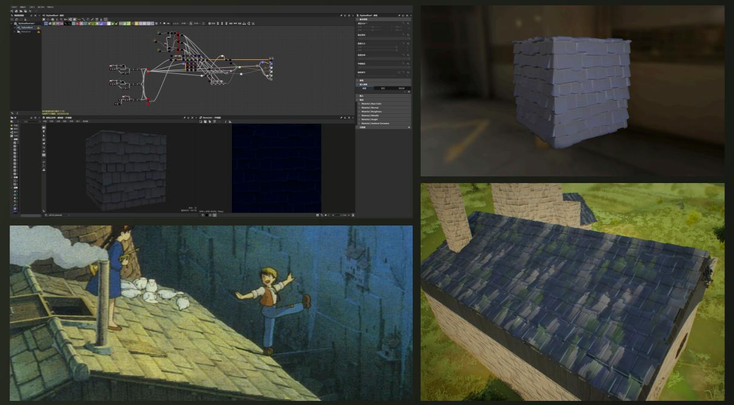
Slope Blue Edge Wear + Flood Fill + Curvature Edge Highlights

Wooden Plank Wall Material

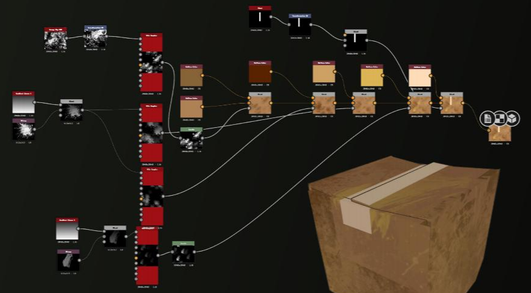
Directional Warp Grain Generation + Tile Generator + Height-Based Moss Mask

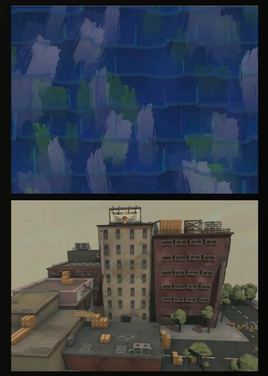
Eaves Material

Tile Sampler + Flood Fil



Finally, I added a few brushstroke touches to enhance the stylized feel—a technique I carried over from experience on another game project.

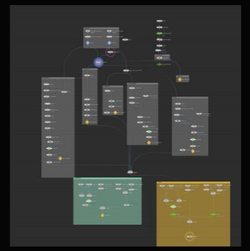


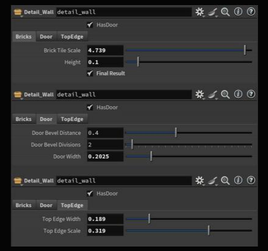


Houdini Procedural Assets

House Brick Wall

*Procedural Feature: Users only need to place two cubes—one to generate a brick wall based on Cube 1's transform, and another to create a door based on Cube 2's transform. When the material texture of the wall is modified, the wall model updates accordingly.**It streamlines the workflow by eliminating the need for manual modeling of complex brick walls.** Parameters: Adjust Bricks: Brick Tile Scale, Brick Height. Adjust Door: Has Door, Door Bevel Distance, Door Bevel Division, Door Width. Adjust Top Edge: Top Edge Width, Top Edge Scale.*





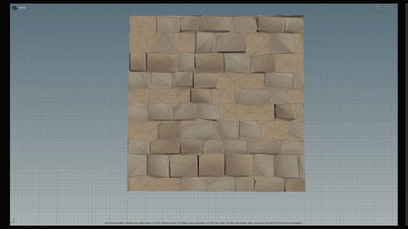
Using House Brick Walls in Unreal Engine

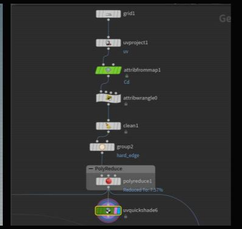
Debugging Position and Shape in Preview Mode

*Click **Is Final Result** to generate the actual model.*

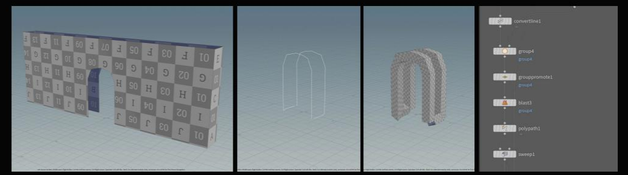
Production Process

*I used the brick material created in Substance Designer, then converted it into a model mesh by applying **Attribute from Map** and **Poly Reduce**.*



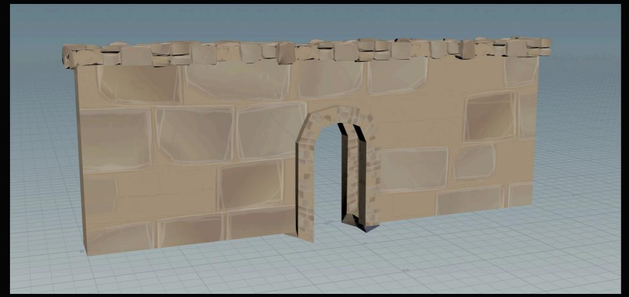


Advanced Workflow:
*I used **Blast** to extract specific faces of a cube, **Boolean** to cut out the door opening, and finally **Sweep** to generate the door frame, top trim, and related structural details.*



Preview Mode Set

I separated performance-intensive nodes using a Switch, allowing users to adjust parameters in Preview Mode first, then toggle the FinalResult button to view the final outcome.

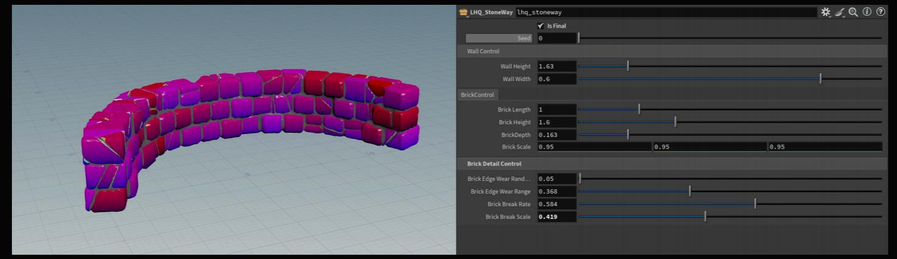


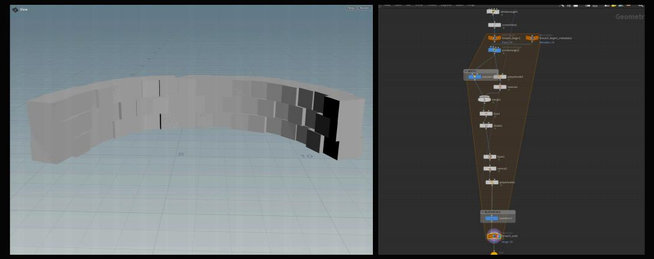
Houdini Procedural Assets

Foreground Stone Wall

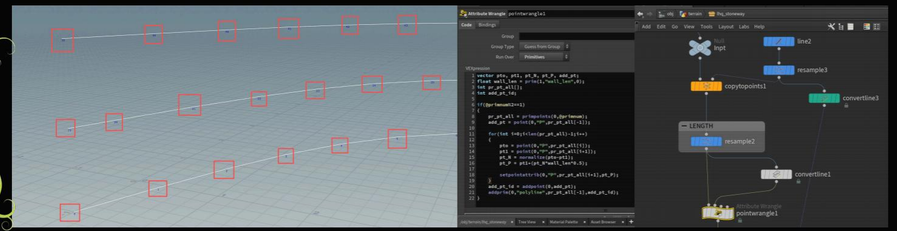
Parameters: The wall's overall length, width, and height can be adjusted, along with the individual stone's length, width, height, and the degree of weathering or breakage.

*Use **For Loop** to modify parameters for each stone.*



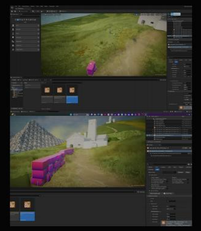


*Key Node: Using **Point Wrangle** to script a pattern: modify the point positions of odd-numbered primitives based on the vector between adjacent points and a given length, then add extra points and polylines. This enables staggered placement of stones.*

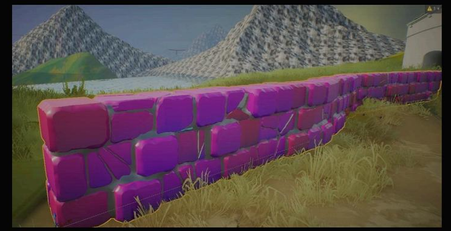


Using Foreground Stone Walls in Unreal Engine

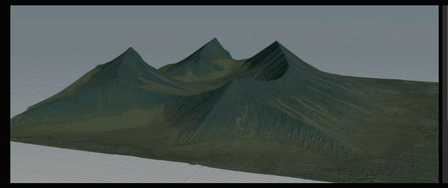
Drag/Adjust Spline

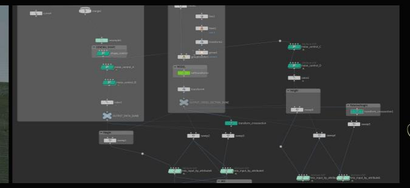


Adjust the desired effects in Preview Mode.



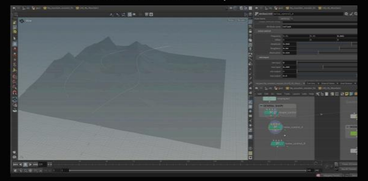
Background Mountain Range



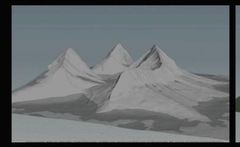


Production Process

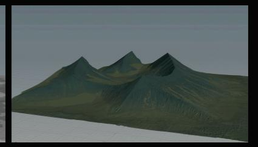
Use various nodes to sculpt the overall form.



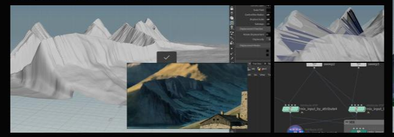
*The mountain range has been enhanced with a **HeightField Erode** node.*



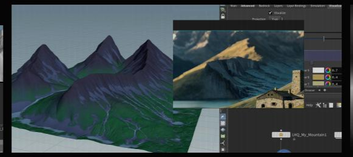
After applying auto-coloration to the mountain range.

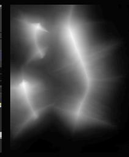


*I utilized **Sweep** combined with **Mix by Attribute** node to create layered mountain forms and blend terrain at varying elevations, roughly sculpting the shape shown below.*



Thought process: Another attempt involved adding elements like rivers, but it felt too realistic and was ultimately not adopted.





*Export as a **HeightMap** for use in Unreal Engine.*

Core Models & UE Materials

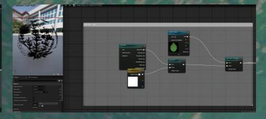
Core Asset – Laputa's Cabin

I created the house model in Maya, as we plan to utilize materials produced in Substance Designer—only UV unwrapping was required.

Stone Wall – Vertex Color Shader

Stone Wall – Vertex Color Material

Runtime Virtual Texturing (RVT) Grass.
This makes the grass blend more naturally with the landscape.



Scene Construction and Iteration

I studied Substance Designer courses and created brick and wood materials for the house. However, they appear too realistic and do not align with my original reference images.

Feedback & Iteration

After working on it, I realized that stylization mainly involves simplifying details—for example, with a brick wall, only basic embossing or highlights are needed.